HARRIS MOHAMED

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EDUCATION

- **University of Illinois at Urbana-Champaign** May 2025
  Master of Science in Computer Science (Machine Learning, Cloud Computing) GPA: 3.74/4.00
- **University of Illinois at Urbana-Champaign** May 2021
  Bachelor of Science in Computer Engineering

Relevant Coursework: Applied Machine Learning, Deep Learning in Healthcare, Databases, Cloud Computing

SELECTED PUBLICATIONS

- **H. Mohamed** [upcoming] AOC EUROPE 2026
  "*AI/ML for Electronic Support: Blind Spectrum Analysis to Characterize Emerging SATCOM Threats*"
- **H. Mohamed** [paper] AMOS 2025
  "Expanding Pattern-of-Life capabilities on Passive Radio Frequency datasets"
- **H. Mohamed** [website] GSAW 2025
  "Methods of Detecting Electromagnetic Interference in Passive Radio Frequency Data"
- **H. Mohamed** [paper] AMOS 2023
  "Monitoring Satellite Pattern-of-Life Changes with Passive Radio Frequency Data"

SELECTED EXPERIENCE

Kratos Defense and Security Solutions Colorado Springs, CO

Artificial Intelligence Engineer October 2025 – Present

- Led first deployment of frontier LLMs (Claude Sonnet 4.5, Claude Code, OpenWebUI) in AWS GovCloud; navigated security and compliance review to enable AI-assisted development across a team of 10+ engineers.
- Built proof-of-concept natural language interface for RF sensor data, enabling engineers to query complex radio frequency measurements and receive both analytical insights and generated data visualizations through conversational AI using Retrieval Augmented Generation (RAG).
- Training 1D CNN and Spectrogram-based neural networks to classify 5G waveform parameters directly from raw IQ samples; achieving 95%+ accuracy using a multi-head mixed-loss architecture in PyTorch.

Space Domain Awareness Software Engineer II October 2021 – October 2025

- Built large-scale data ingestion system (Dagster + PostgreSQL) processing data from 20 globally distributed edge sensor nodes into a 65M+ row analytical data lake with high-throughput, fault-tolerant ingestion.

NVIDIA Remote

Computer Architecture Intern Summer 2021

- Contributed to GPU simulator architecture design and verification on next-generation hardware; gained deep understanding of GPU compute pipelines, memory hierarchies, and accelerator design.

SELECTED PROJECTS

Atlas – NanoClaw fork for self-hosted Agent AI Assistant [GitHub] September 2025 - Ongoing

- Built self-hosted agentic AI system using Claude Agent SDK with multi-agent orchestration; implements tool use (GitHub API, Brave Search, credential management via onecli) and verification loops where specialized agents generate, fact-check, and validate reports against format specifications.

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, Reinforcement Learning (DQN, PPO)

Infrastructure: AWS (Bedrock, EC2, GovCloud), Docker, vLLM, Ollama, CUDA, Dagster, InfluxDB, PostgreSQL, MongoDB

Languages: Python, C++, CUDA C++, JavaScript/TypeScript, SQL

Tools: Git, Linux, Jupyter, LaTeX, Grafana